UKAPA7 Combined Comparison Report

This report combines the UKAPA7 difference-distribution analysis and the side-B procedural interpretation into one document.

It compares two full-wire tension measurement campaigns for UKAPA7 layer G:

- UK factory measurements in `apa_uk7g.json` on December 10, 2023
- Chicago factory measurements in `data/tension_summaries/tension_summary_UKAPA7_G.csv` on March 11, 2026

Residual definition throughout: `Chicago - UK`.

Executive Summary

- Side A has the expected sign: Chicago tensions are lower than UK tensions on average, with mean residual `-0.376 N`.
- The operational explanation is that side B was measured in the wrong direction. The correct procedure is to start from the low slot on side A and from the high slot on side B.
- If side B was instead measured starting from the low slot, the B-side wire order would be reversed.
- This revised report therefore uses only the reversed-B interpretation and does not apply any additional index correction.

Physical Expectation

The expected physical change from the UK campaign to the Chicago campaign is a roughly constant negative offset, since the later Chicago tensions should tend to be lower.

Both measurement campaigns are noisy. If the single-measurement uncertainty is about `0.25 N` to `0.5 N`, then the residual scatter expected from two independent noisy measurements is roughly `0.35 N` to `0.71 N`.

The observed residual widths are around `0.41 N` to `0.43 N`, which is compatible with that noise scale. The main issue is therefore how side-B wires are paired between the two campaigns.

Source Artifacts

- Raw side-A comparison:

  `data/tension_plots/tension_raw_A_UKAPA7_G.png`
- Raw side-B comparison, reversed indexing:
  `data/tension_plots/tension_raw_B_reversed_UKAPA7_G.png`
- A-side versus reversed-B residuals:
  `data/tension_plots/tension_a_vs_reversed_b_UKAPA7_G.png`

Raw Data Plots

These plots show the original UK and Chicago tensions directly. Each plot uses
the same format:

- top panel: per-wire scatter with rolling trendlines
- bottom panel: overlaid histograms
- statistics shown directly on the plot

Side A Raw Tensions

![UKAPA7 raw side A](../data/tension_plots/tension_raw_A_UKAPA7_G.png){ width=100% }

Side B Raw Tensions, Reversed Indexing

![UKAPA7 raw side B reversed](../data/tension_plots/tension_raw_B_reversed_UKAPA7_G.png){ width=100% }

Raw Data Summary

Comparison	Count	Chicago Mean (N)	UK Mean (N)	Corr	Mean Diff (N)	Std Diff (N)	MAE (N)
Side A	481	5.761	6.137	0.227	-0.376	0.410	0.484
Side B reversed	481	5.849	6.048	0.168	-0.199	0.435	0.391

Interpretation

- Side A shows a stable negative offset between the UK and Chicago campaigns.
- The reversed-B plot is included because it matches the procedural hypothesis:
  if Chicago side B was measured from the low slot instead of the high slot,
  then the B-side ordering would be reversed.

A Side Versus Reversed B Side

This figure compares:

- side A under the normal indexing convention

- side B under the reversed-index hypothesis implied by the measurement procedure

A Versus Reversed B Statistics

Series	Count	Mean (N)	Median (N)	Std (N)	MAE (N)	Max Abs (N)	Negative Fraction
A baseline	481	-0.3762	-0.4269	0.4103	0.4843	1.6307	0.8441
B reversed	481	-0.1994	-0.2385	0.4346	0.3914	1.5501	0.6840

Interpretation

- Under the reversed-B interpretation, side B remains less negative than side A and has a somewhat broader residual distribution.
- That means the reversal alone does not make side B look identical to side A.
- Even so, reversal is the right procedural hypothesis to visualize, because it reflects what would happen if the Chicago side-B scan started from the wrong end.

What Likely Happened Operationally

The likely procedural sequence is:

- side A was measured correctly from the low slot
- side B should have been measured from the high slot
- side B was instead measured from the low slot

That would reverse the side-B ordering relative to the intended wire numbering.

This report therefore presents side B under that reversed interpretation and uses it as the operational explanation for the mismatch.

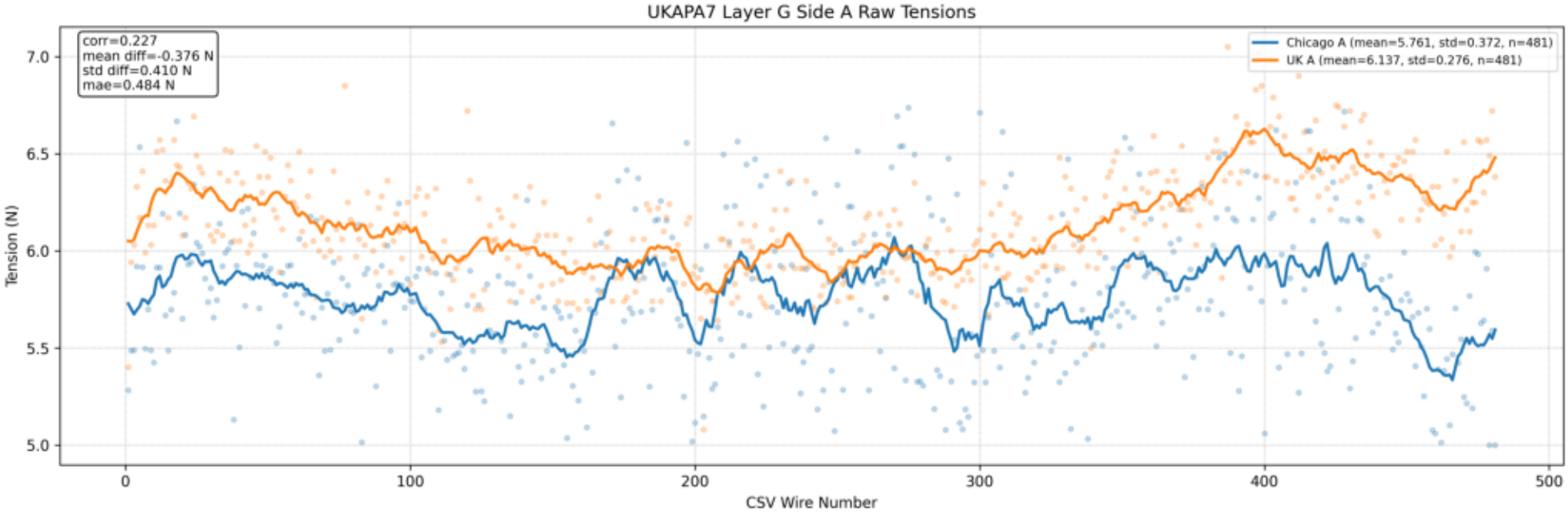
Conclusion

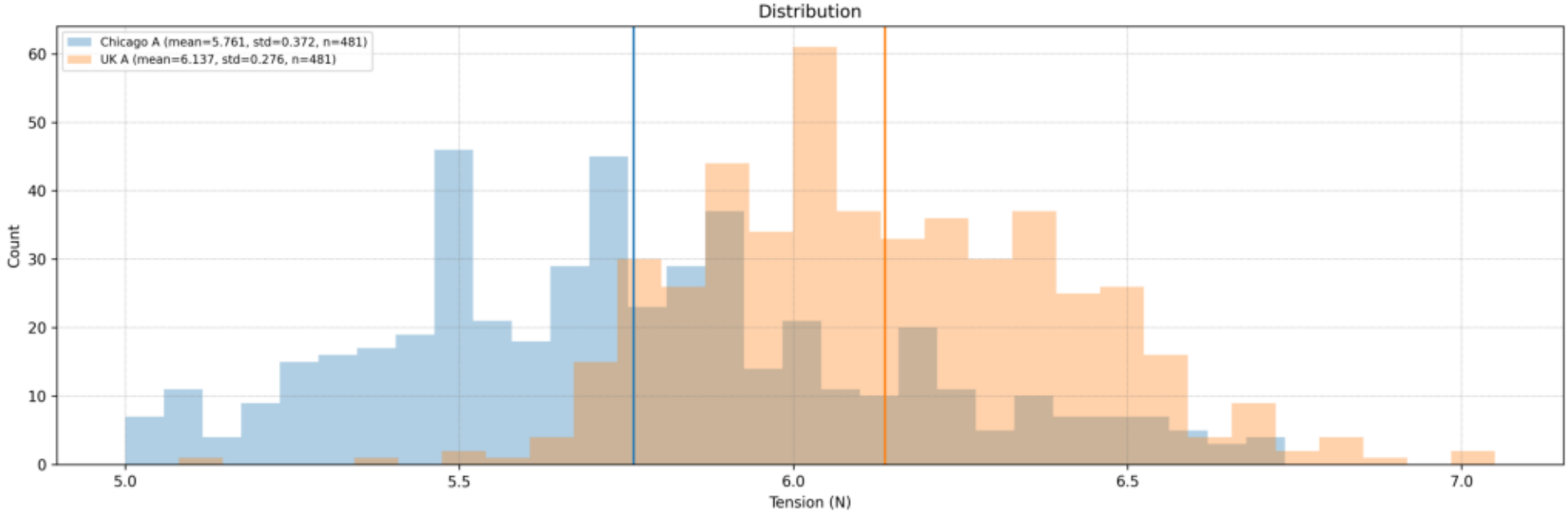
The combined evidence supports the following interpretation:

- the overall UK-to-Chicago change has the expected negative sign
- side A is broadly consistent with that picture
- the most relevant procedural explanation is that side B was measured in the reverse direction, starting from the low slot instead of the high slot

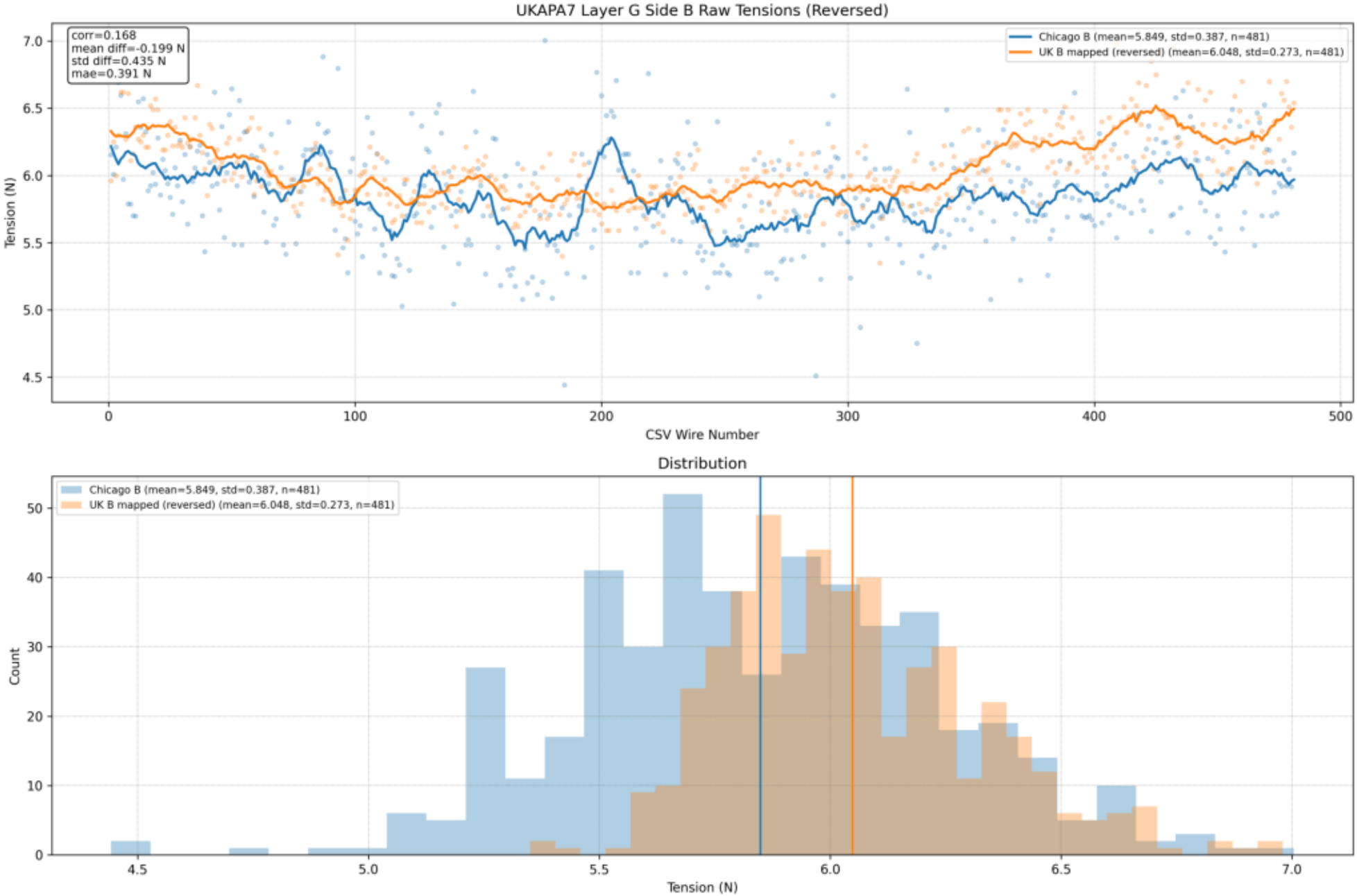
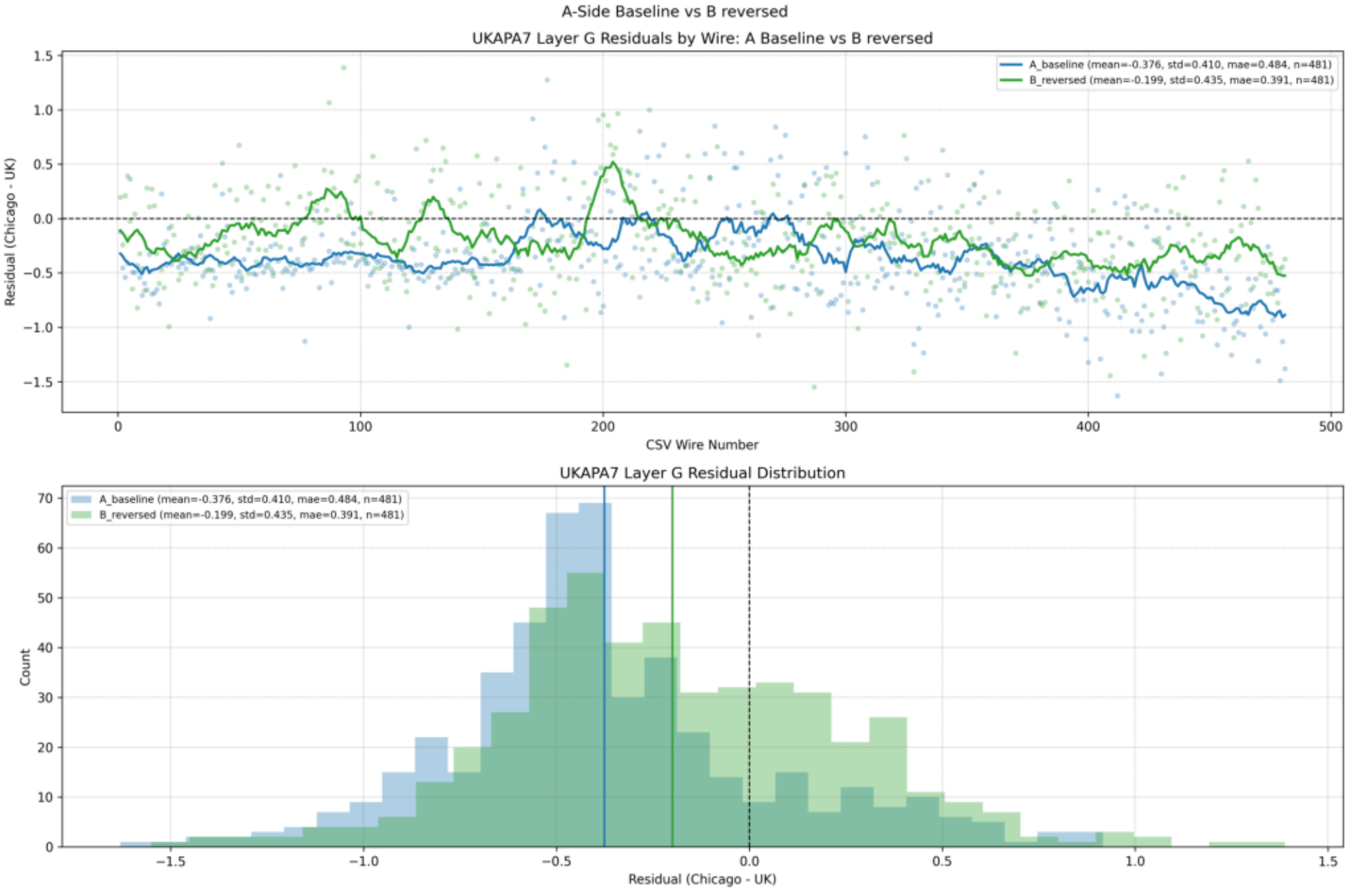
In practical terms, the operational fix is straightforward: measure side A from the low slot and side B from the high slot, and verify that the side-B wire order in the output follows that convention.

tension_raw_A_UKAPA7_G.png





tension_raw_B_reversed_UKAPA7_G.png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